Download Unity Files

- From the **3D_Models_Server** google drive, navigate to the **Unity_M** folder.
- Download **Unity_Demo_M.zip** file

Installing Unity and Softwares

- Download Unity/Unity Engine
  - https://unity.com/download
- Install appropriate editor version
  - Project runs on **2021.3.1f1**
- Install Visual Studio (the purple one) (or use an IDE of your choice)
  - https://visualstudio.microsoft.com/

Main Assets:

(Note: these assets are already set-up in the project. Information is here for your convenience.)

- Shackleton_Ridge_300x300_v4_/**Lunar_Floor**
  - Lunar Terrain

- **Unity_Viper_Frame**
  - Viper Rover (real)

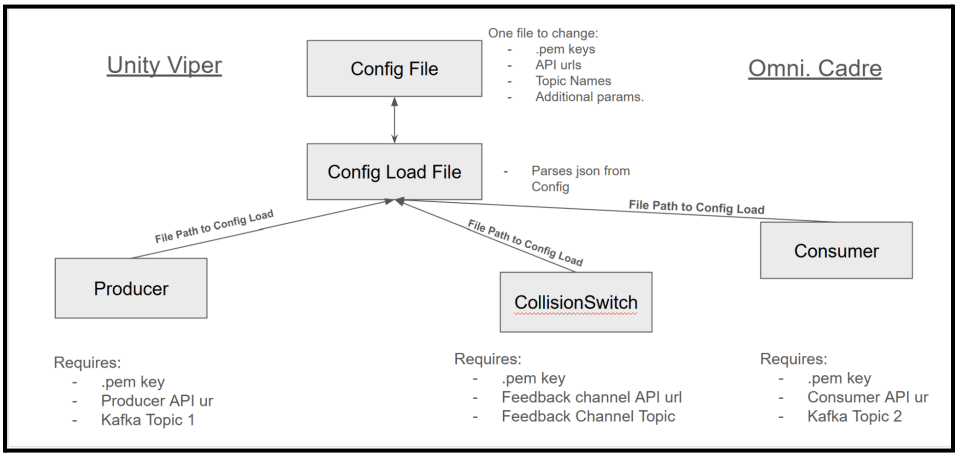
- **Omni_CADRE (1)**
  - Cadres Rover (Ghost)

- **Waypoints (1)**
  - Waypoints that Unity_Viper_Frame follows
  - Can be manually moved

Scripts:

(Note: these scripts are already set-up in the project. Information is here for your convenience.)

- Configuration Scripts
  - Assets\Scripts**HsmlConfigLoader.cs**
    - File used by all scripts that need to access hsml_config.json
      - This is so that you only need to change secrets/data once rather than in multiple scripts
      - Also allows cross-platform (windows, macOS, etc.)
    - Parses hsml_config.json so that each script doesn't have to do its own parsing.

  - Assets\StreamingAssets**hsml_config.json**
    - Essentially where you store all your secrets/data you want to send
      - Simply put the correct API URL, path to .pem keys, any other data, etc.
    - Visual Sample of how it works in Unity:
    - 

- Scripts attached to Unity_Viper_Frame (Real)
  - Assets/Scripts/MovementScripts/**viperController.cs**
    - Script that enables user to manually control Viper rover in Unity
      - Note: using this script requires any waypoint scripts to be disabled

  - Assets/Scripts/HSMLapi/**unityApiProducer.cs**
    - Script that enables Viper to "produce" messages

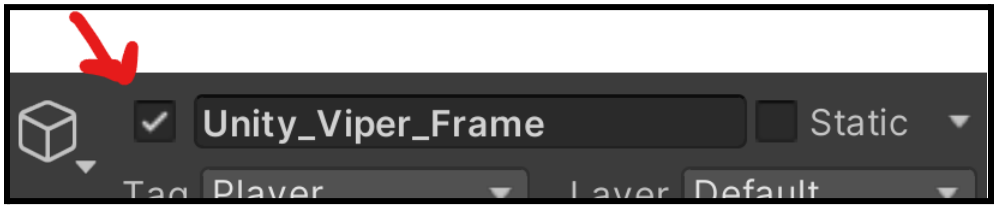
- Assets/**realisticwaypoint.cs**
    - Script that tells Viper to follow a set of waypoints, such as **Waypoints (1)**

- Scripts attached to Omni_Cadre (Ghost)
    - Assets/Scripts/HSMLapi/**omniApiConsumer.cs**
        - Script that enables Cadre to “consume” messages

    - Assets/Scripts/NEWNEW/**ActivateRover.cs**
        - Script that triggers on-collision; Disables omniApiConsumer.cs and enables switchOmniApiConsumer.cs

    - Assets/Scripts/NEWNEW/**switchOmniApiConsumer.cs**
        - Script that enables Cadre to “produce” messages
            - Note: This script should be disabled at start.

Running Simulation:

1. **Ensure Main Assets are enabled**
    a. (Enabled in Unity means asset is checkmarked; Sample image below)
    b. 

2. **Ensure appropriate scripts are enabled unless otherwise specified**
3. **Press play**
    a. Note: it may be easier to view the simulation through the scene view; Game view requires cameras